Homeostasis During Fasting. II. Hormone Substrate Differences Between Men and Women

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ABSTRACT. Ten normal men and 9 normal women were fasted for 72 hr. Four men, matching 4 of the women in weight and age, were also fasted a second time while receiving ethinylestradiol (0.1 mg twice daily). Five of the women were fasted in both the preovulatory and postovulatory phases of the menstrual cycle.

Premenopausal women had significantly lower concentrations of glucose during fasting than men. The mean lowest glucose concentration in plasma in the male group during the 3 days of starvation was 66.4 ± 2.9 mg/100 ml ($\bar{X}$ ± SEM). The corresponding figure in females was 47.8 ± 2.9 mg/100 ml ($p < 0.01$). Associated with the greater decrease of plasma glucose in women was a significantly greater and earlier increase of plasma immunoreactive glucagon. Mean peak glucagon concentrations in women and men were, respectively, 230.0 ± 36.1 μμg/ml and 121.8 ± 17.9 μμg/ml ($p < 0.01$). These differences of glucagon were, likewise, significant when expressed in terms of per cent increase from basal (170 ± 25% *vs.* 56 ± 20%, $p < 0.01$).

There was no significant difference in plasma insulin concentrations during the fast, but plasma free fatty acid (FFA) concentration rose significantly higher in the women than in the men (mean peak concentration: women, 2.25 ± 0.204 mmole/l *vs* men, 1.429 ± 0.149 mmole/l, $p < 0.01$). An attempt to correlate the hormone-substrate response with (a) body weight of the subjects, (b) treatment or nontreatment with estrogen or (c) phases of the menstrual cycle was unsuccessful.

The data obtained from these fasts supports a different hormone-fuel pattern of response for men and women. The data are not consistent with a single, primary hormone of fasting. (*J Clin Endocrinol Metab* **37**: 698, 1973)

THE regulation of substrate production during starvation has been the subject of intense investigation (1–3). Although insulin has been suggested as a primary hormone of fasting, (1) we were able to show in previous studies that plasma glucose concentrations could not be maintained adequately in the chronic absence of human growth hormone (hGH) (2).

The present study will emphasize three points not appreciated previously concerning the hormone-substrate changes during fasting: (a) that premenopausal women do not maintain plasma glucose concentration during a fast as do men of a comparable age, (b) that there is a greater increase of plasma glucagon in women and, likewise, (c) a greater increase in plasma free fatty acids (FFA).

Materials and Methods

Prior to admission, all subjects ingested a diet containing a minimum of 150 g of carbyhodrate and 1 g/kg of body weight of protein for at least 2 weeks. Informed consent was obtained from 9 women and 10 men.[1] Four men were closely matched for body weight and age with 4 of the women. The men were fasted twice, both with and without ethinylestradiol, 0.1 mg twice daily. Five of the women were fasted during the preovulatory and postovulatory phases of the menstrual cycle. No subject had CHO intolerance and hepatic screening tests were normal. No subject was receiving any type of medication.

Each 3-day fast was preceded by a 2-day control period. Samples were obtained at 8 AM and 8 PM daily. Four ml of blood were added immediately to a chilled tube containing 0.4 ml of Trasylol (1000 U/ml). Twenty ml of blood were collected in a heparinized syringe; both aliquots were centrifuged immediately and stored at −20C until analysis. The samples were analyzed for glucose, FFA, insulin, HGH and glucagon concentrations. Glucose was measured in triplicate by a glucose oxidase method (4). The coefficient of variation between triplicates in a given assay was 1.5% and between assays, 2.5%. FFA were measured in duplicate by the method of Trout, Estes and Friedberg using palmitic acid standards and a 4 point standard curve (5). Serum immunoreactive insulin, HGH

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[1] Within 10% of ideal body weight according to New York Metropolitan Tables.

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and glucagon were assayed by modifications of the methods of Berson and Yalow, and Roth, Glick, Yalow and Berson, respectively (6,7). HGH samples were checked in duplicate with the maximum variation between duplicates being 0.5 ng/ml in the 0 to 5 range, and 1.0 ng/ml in the 5 to 15 range. To obtain maximum accuracy within the range of normal basal values of insulin, samples were run in quadruplicate, the final reported value being the mean of 4 determinations. Maximum variation in the basal range (0 to 15) was 1.0 μU/ml. Glucagon was measured using a charcoal separatory method and antiserum for pancreatic glucagon (30 K) which was obtained from Dr. Roger Unger. For glucagon measurements, samples were run in triplicate with appropriate control tubes for each sample. To negate small differences between assays with glucagon, the values used in this report were obtained by reassaying all samples in a single assay.

The reference hormones used for each assay were as follows: HGH, HS 1394; Glucagon, GLF 599A; and for Insulin, PJ 5682 (Lilly).

Results

A. *Men/women pattern.* Fig. 1 depicts the mean plasma glucose, FFA, glucagon and insulin concentrations of normal men and women at 12-hr intervals during the control period and 72 hr of fasting. Sample 1 was obtained after a 12-hr fast while patients were receiving the standard diet. With prolonged fasting, plasma glucose and insulin concentrations decreased in both men and women, reaching a plateau on the second or third day of the fast.

There was a striking sex-based difference in the pattern of glucose change. Basal plasma glucose concentrations were similar, being 89.0 ± 2.5 mg/100 ml in men and 86.3 ± 3.2 mg/100 ml in women. In women, plasma glucose decreased to 52.5 ± 3.4 mg/100 ml at 60 hr *vs* 67.5 ± 2.0 mg/100 ml in men. The difference between men and women while fasting was significant at the 1% level. The mean lowest glucose level in the men was 66.4 ± 2.9 mg/100 ml and in the female group, 47.8 ± 2.9 mg/100 ml ($p < 0.01$).

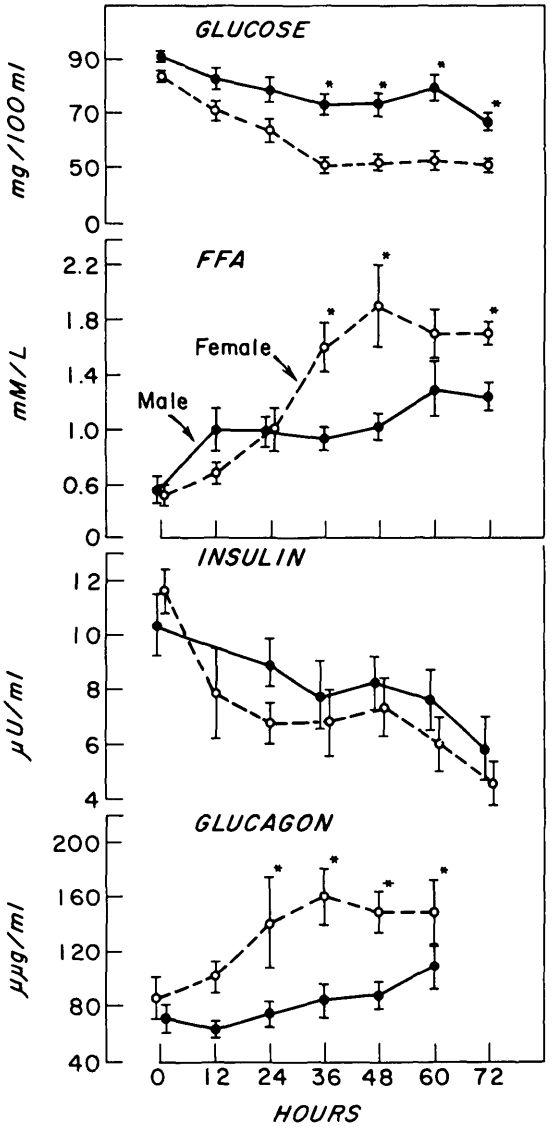

FIG. 1. Plasma concentrations of glucose, FFA, insulin and glucagon are shown for men and women at 12-hr intervals during 3 days of fasting. * Indicates values differ with a p value of <0.01. Points are mean ± SEM.

Basal glucagon concentrations in women and men were 85.1 ± 15.4 pg/ml *vs* 75.0 ± 7.3 pg/ml, respectively. At 36, 48 and 60 hr of the fast, plasma glucagon concentrations were significantly higher in women than in men ($p < 0.01$). Actual glucagon concentrations at all time intervals are illustrated in Fig. 1 and given in Table 1. The corre-

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TABLE 1. Glucose and glucagon concentrations in plasma: group comparisons with 72 hours of fasting

	Glucose mg/100 ml		Glucagon pg/ml	
	Mean basal conc.	Mean lowest conc.	Mean basal conc.	Mean highest conc.
A. Normal males (9)	87.3 ± 3.5	66.4 ± 2.8	75.0 ± 7.3	122.0 ± 17.9
Normal females (9)	86.3 ± 3.2	47.8 ± 2.9	85.1 ± 15.4	230.0 ± 26.1
	p NS	p < .01	p NS	p < .01
B. Light weight males (4)	90.1 ± 5.0	65.2 ± 4.9	70.0 ± 8.2	132.0 ± 16.0
"Matched" females (4)	86.3 ± 3.2	47.0 ± 4.8	87.0 ± 9.5	248.0 ± 17.3
	p NS	p < .01	p NS	p < .01
C. Females—preovulatory* (5)	88.0 ± 4.0	51.0 ± 3.0	—	278.0 ± 16.0
Females—postovulatory (5)	86.3 ± 3.2	47.0 ± 6.9	83.0 ± 12.6	290.0 ± 23.0
	p NS	p NS	—	p NS
D. Untreated males (4)	90.0 ± 6.0	71.0 ± 4.9	70.0 ± 8.4	109.0 ± 14.2
Treated with ethinylestradiol (4)	87.0 ± 3.2	68.0 ± 5.0	83.0 ± 10.0	116.0 ± 7.8
	p NS	p NS	p NS	p NS

* First three subjects studied are shown in Fig. 2.

lation coefficient for glucagon and glucose for men was 0.64 ($p < 0.01$) and for women was 0.51 ($p < .01$). The slope of the regression line for men was −1.48 *vs* −2.20 for women.[2]

Prefasting plasma insulin levels were similar in men and women; 11.9 ± 2.3 μU/ml and 11.8 ± 1.7 μU/ml, respectively. A similar decrease of insulin occurred during the fast in each of these groups.

Plasma HGH levels were extremely variable. Mean figures at all time intervals studied did not differ between groups, nor within any given group.

B. *Effect of the menstrual cycle and body size on glucose-glucagon patterns.* Women were fasted in both the preovulatory and postovulatory phase. The former study was conducted 4 to 5 days after the last day of menstrual bleeding; the latter, 15 to 17 days after the last day of menstrual bleeding. Changes of plasma glucose and glucagon were similar in each study period (Fig. 2, Table 1).

Results of fasting 4 men selected to match 4 women in weight and age are illustrated in Fig. 3. Plasma concentrations of glucose did not decrease to a greater degree in these men compared with the initial group of unselected men of greater body weight. Plasma glucose concentrations were significantly higher, however, in these men than in the weight-matched women. Thirty-one of 36 values of plasma glucose concentrations after the first 24 hr of fasting were less than 60 mg/100 ml in women contrasting with

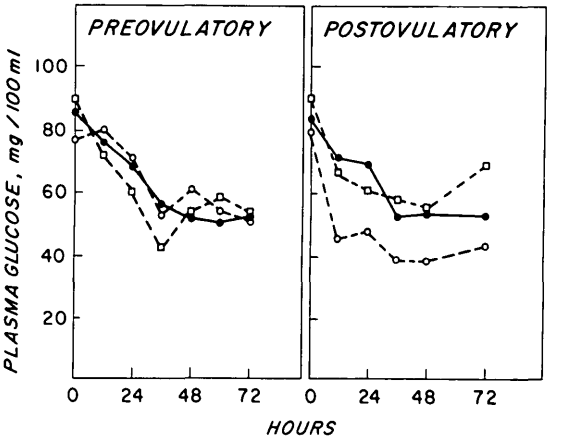

FIG. 2. The glucose concentrations in plasma are shown for 3 women while fasting in the preovulatory and postovulatory phases of the menstrual cycle.

[2] Regression formula: for men, $y = -1.48x + 192$, degrees of freedom = 43; for women, $y = -2.20x + 285$, degrees of freedom = 43.

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only 5 of 36 values less than 60 mg/100 ml in men ($p < 0.01$). When these men were treated with 0.1 mg of ethinylestradiol twice daily for 4 days prior to and during a similar fast, the glucose-glucagon pattern in plasma did not change. Detailed data for all studies are given in Table 1.

C. *Plasma FFA*. Considerable variation was noted in plasma FFA concentrations in each group and in each individual. Absolute values (mmole/l) were higher in women than in men during the fast. The actual difference was significant at 12, 36 and 72 hr of the fast. The mean maximal FFA response recorded in women while fasting was 2.258 ± 0.204 mmole/l *vs* 1.55 ± 0.110 mmole/l in the men ($p < 0.01$).

Since there was considerable variation of the basal plasma FFA concentrations in both groups, FFA were, likewise, calculated in terms of percent increase over basal. The mean maximal female response was still significantly greater than in the male response (480% ± 81% *vs* 314% ± 43%), but the differences between the 2 groups were less impressive.

Discussion

Carbohydrate stores provide only a small fraction of the body's fuel requirements during any fast exceeding 24 hr. The major source of emergency fuel is then provided by amino acids mobilized from protein and FFA from adipose tissue (8–10). Several hormones have been suggested as possible agents controlling the release and usage of these substrates. Although early studies showed an increase of HGH during a fast (3), this finding was often discounted due to the sporadic nature of the HGH response and, more importantly, to its lack of correlation with any measurable metabolic parameters (1,2). Since the total 24-hr secretory pattern of HGH was not assessed in these studies, this latter conclusion may or may not be justified.

Insulin was later offered as the hormone primarily responsible for control of substrate during starvation (1). However, when

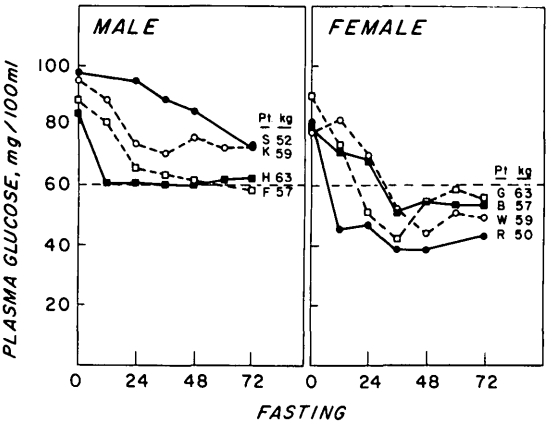

FIG. 3. Glucose responses of 4 age and weight-matched men and women are shown at 12-hr intervals during 72 hr of fasting. Weights of each subject are indicated.

we fasted hGH-deficient dwarfs, plasma glucose concentrations were not adequately maintained (2) suggesting that the homeostatic mechanism in starvation was more complex than initially considered. The working hypothesis which evolved in our laboratory to explain this data was as follows: First, insulin, as postulated by Cahill, was thought to act as a fulcrum regulating glucose/FFA concentration in plasma, but, second, the setting of this mechanism was, in some manner, HGH dependent. To begin clarifying this data, studies were first designed to investigate several alternative explanations.

At the onset, we tested the adequacy of our control group, since in all fasting studies, including our own, control subjects were invariably men. This enlargement of the control group to include men and women resulted in the principal finding reported in this paper: a prominent sex variation in glucose control.

The differences of glucose-glucagon responses to fasting between normal men and women were striking and unequivocal. Premenopausal women had greater FFA mobilization than men in the face of lower glucose concentrations during a fast. As indicated in the previous section, the greater glucagon response of women appeared to correlate with this greater decrease of plasma glucose

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concentration. Insulin concentrations decreased in the women, but not to a lower level than in the men.

It would be attractive to postulate that estrogen is responsible for the major differences noted between normal men and women. One might hypothesize that since estrogen counters many effects of HGH, normal women would behave as do HGH-deficient subjects (11–13). The data, however, do not support this conclusion. The same women fasted at different periods of the menstrual cycle had similar changes of glucose and glucagon. Furthermore, men treated with ethinylestradiol did not exhibit a metabolic pattern comparable to women matched in age and weight.

A more likely explanation for the results relates to differences in body proportion. Although we matched precisely four subjects of each sex for body weight and age, a difference in adipose tissue to muscle mass ratio could still exist (14). A reduced muscle compartment in women relative to men would result, theoretically, in a dimunition of plasma amino acids for conversion to glucose plus reduced uptake of FFA by this tissue compartment. A larger adipose tissue compartment in women should yield a higher plasma concentration of FFA. This explanation would explain nicely the difference observed in glucose levels and, in addition, would be consistent with the observation that although decreases of plasma insulin were similar, increases of plasma FFA concentrations were disparate. Whether glucagon is, in any way, responsible for the greater increase of FFA in women cannot be determined from our data, nor do we know if greater glucagon concentrations will modify glucose concentrations during a longer fast.

Additional studies quantifying adipose and muscle compartments of the unmatched men and women remain to be performed, but the data seems unequivocal in two separate respects: there is a greater decrease of plasma glucose in women than in men during fasting and probably, as a result, a greater increase of plasma glucagon. Considering the data in more general terms, it seems obvious that current thought regarding the hormonal control of fasting has been oversimplified.

Maintenance of adequate substrate during starvation probably involves several hormones. The relative concentrations of each in regard to the other may also be of importance. The hormone-fuel pattern in plasma probably reflects not merely endocrine secretion, but the tissue masses reacting to this secretion.

Acknowledgments

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